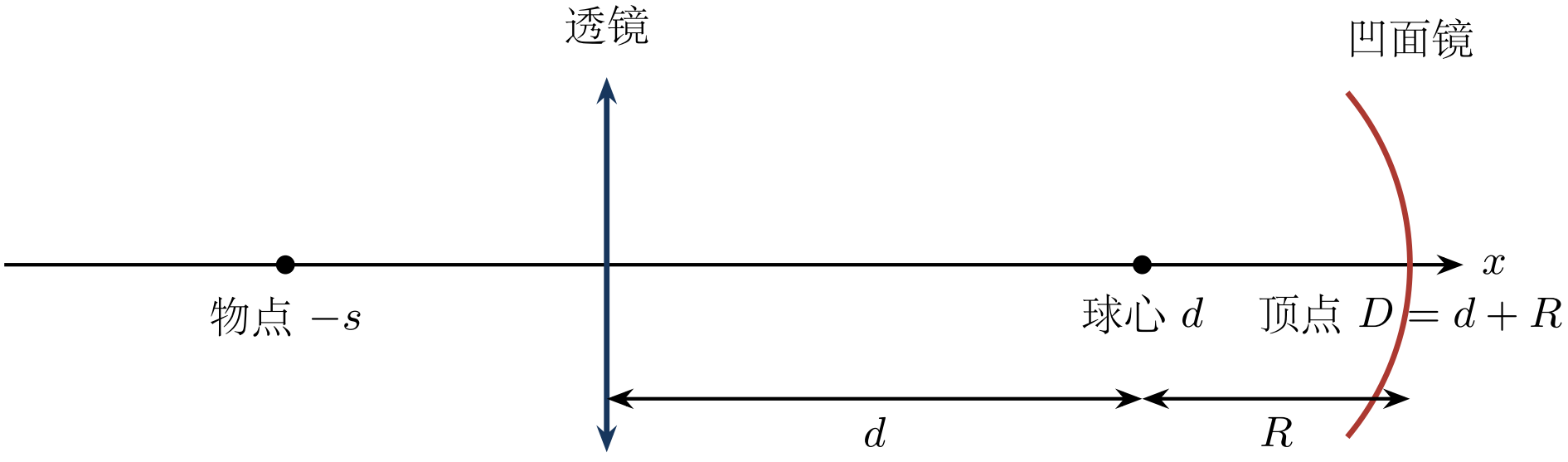
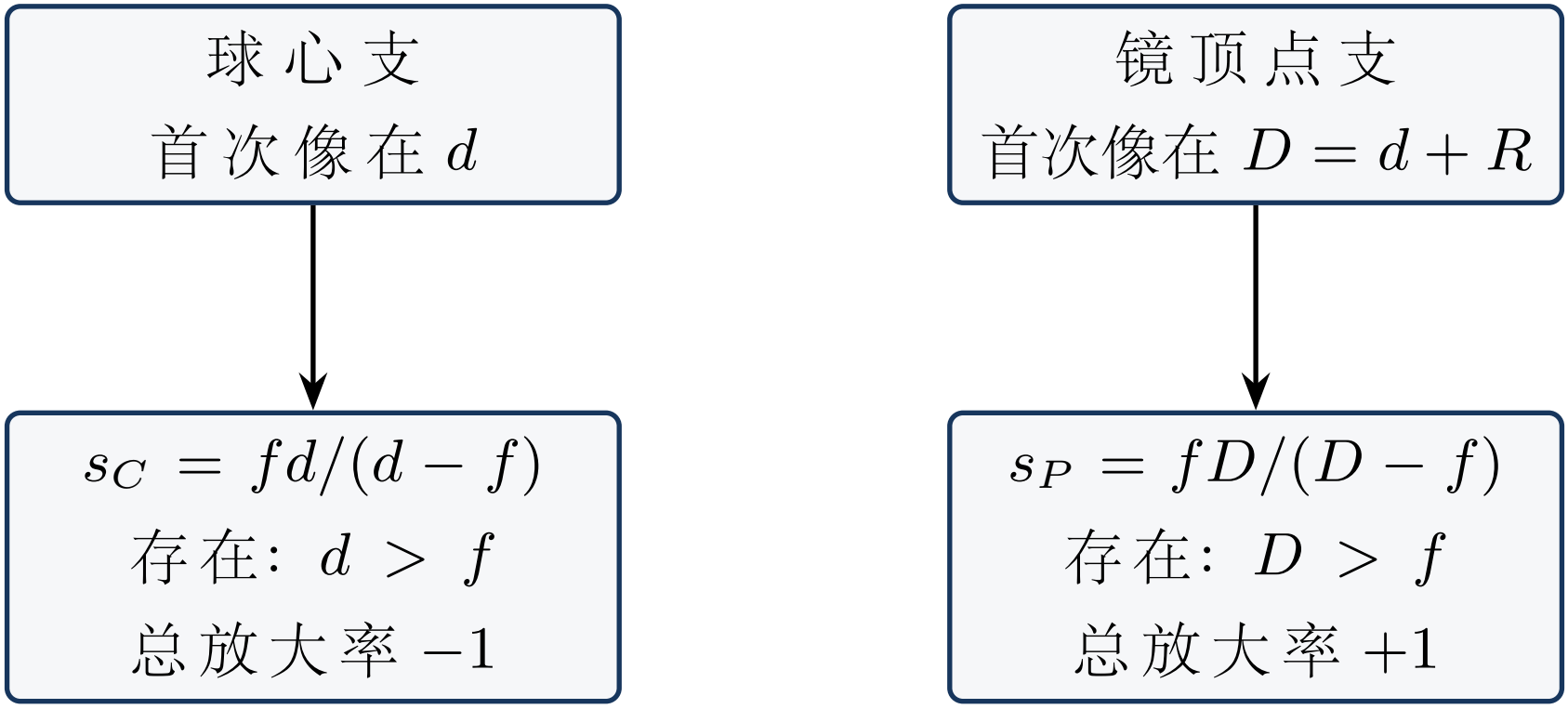


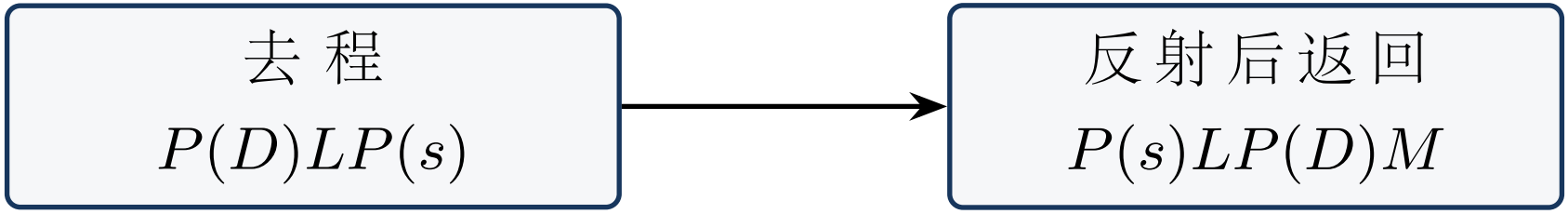

物点在透镜左侧；此图取球心在右侧 $d \geq 0$

镜口面向透镜，返回光第二次经过透镜




仅要求轴上物点自共轭时两支均可

完整图案同位正立还须选择顶点支

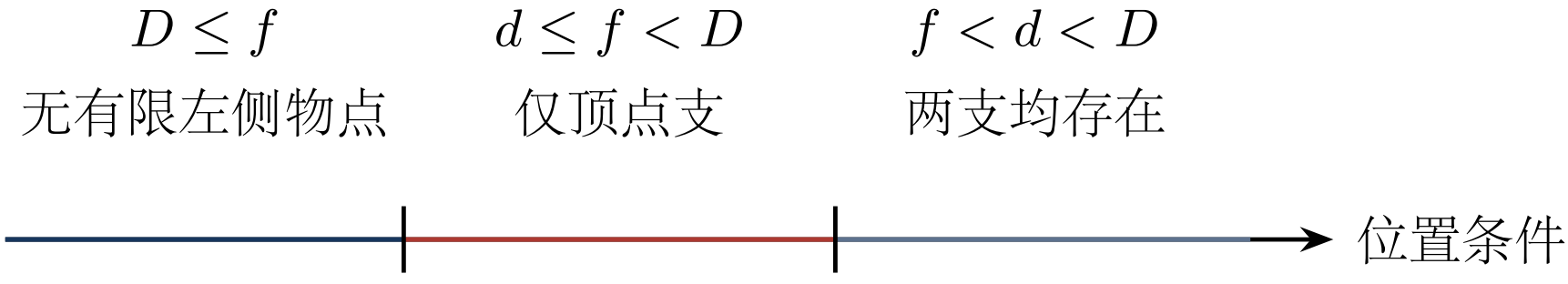



$$T = P(s)LP(D)MP(D)LP(s)$$

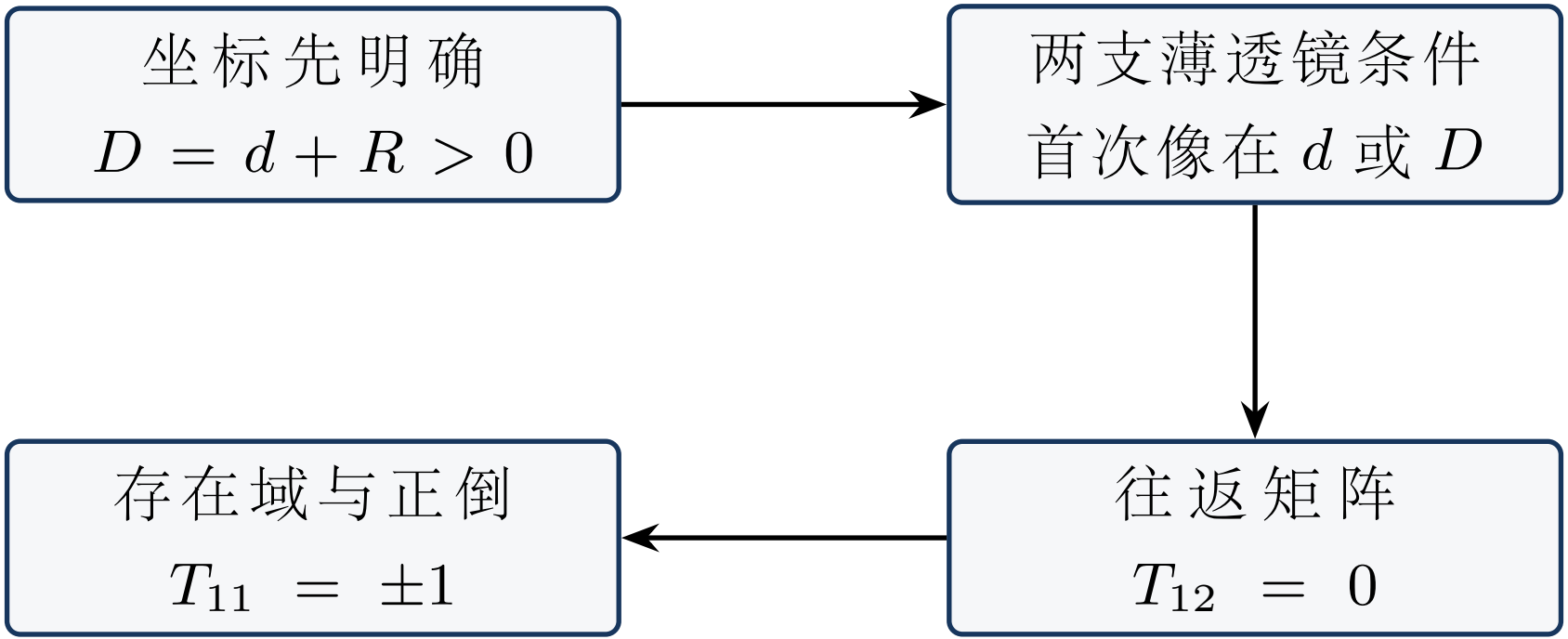
$$P(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix},\ L = \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}$$

$$M = \begin{pmatrix} 1 & 0 \\ -2/R & 1 \end{pmatrix}$$

自共轭条件：$T_{12} = 0$，总放大率为 $T_{11}$




$d = f$ 时球心支移到无穷远；$D = f$ 时顶点支也移到无穷远




$s_C > s_P$ 在两支都存在时成立